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ORIGINAL SCIENTIFIC REPORT

The Prognostic Impact of Extent of Vascular Invasion in Follicular Thyroid Carcinoma

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Abstract

Background Encapsulated angioinvasive follicular thyroid carcinoma (EAFTC) is associated with an increased risk of distant metastasis and reduced survival compared to minimally invasive follicular thyroid carcinoma (MIFTC). There is controversy regarding the extent of surgery and adjuvant radioactive iodine therapy for angioinvasive follicular thyroid carcinoma when stratified by number of foci of angioinvasion.

Methods All follicular thyroid carcinoma cases from 1990–2018 were identified from a thyroid cancer database. Primary outcomes were distant metastasis-free survival (DMFS) and disease-specific survival (DSS) with factors of interest being age, gender, tumour size, treatment, foci of angioinvasion and histological subtype.

Results A total of 292 cases were identified; 139 MIFTC, 141 EAFTC and 12 widely invasive follicular thyroid carcinoma (WIFTC). Over a follow-up period of 6.25 years, DMFS was significantly reduced ($p < 0.001$) with 14.2% (EAFTC) and 50% of WIFTC developing metastasis. The risk of metastasis in EAFTC with ≥ 4 foci of angioinvasion was 31.7% (HR = 5.89, $p = 0.004$), 6.3% for EAFTC with < 4 foci of angioinvasion (HR = 1.74, $p = 0.47$), compared to 3.6% MIFTC. Age ≥ 50 years (HR = 4.24, $p = 0.005$) and tumour size (HR = 1.27, $p = 0.014$) were significantly associated with increased risk of distant metastasis. DSS was reduced significantly ($p < 0.001$), with 7.8% EAFTC patients dying of disease. For EAFTC patients, DSS was 96.8% for < 4 foci and 82.6% for ≥ 4 foci of angioinvasion ($p = 0.003$).

Conclusion EAFTC is at increased risk of distant metastasis related to the extent of angioinvasion. Tumours with < 4 foci of angioinvasion should be considered for a total thyroidectomy, particularly in older patients.

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Abbreviations

FTC	Follicular thyroid carcinoma
EAFTC	Encapsulated angioinvasive follicular thyroid carcinoma
MIFTC	Minimally invasive follicular thyroid carcinoma
WIFTC	Widely invasive follicular thyroid carcinoma
DMFS	Distant metastasis-free survival
DSS	Disease-specific survival
RAI	Radioactive iodine

Introduction

Follicular thyroid carcinoma (FTC) is the second most prevalent type of differentiated thyroid carcinoma and accounts for 10–15% of all thyroid cancer presentations [1, 2]. FTC occurs predominantly in adults between 45 and 50 years of age, and it is estimated that up to 10% present with distant metastasis [1]. The presence of distant metastasis correlates directly with disease-specific survival from follicular thyroid carcinoma [3]. Key histological criteria distinguishing FTC from a benign follicular nodule are based on the presence of capsular and/or angioinvasion [4]. Historically, FTC was divided into being either minimally invasive or widely invasive tumours [5]. However, FTC is now classified into three distinct categories; minimally invasive follicular thyroid carcinoma (MIFTC) negative for vascular space invasion, encapsulated angioinvasive follicular thyroid carcinoma (EAFTC) positive for vascular space invasion with or without capsular invasion and widely invasive follicular thyroid carcinoma (WIFTC) with extensive capsular invasion with or without vascular invasion in order of increasing risk of distant metastasis and reduced survival [4]. Over the last decade, angioinvasive follicular thyroid carcinomas (EAFTC) have gained recognition as a distinct higher risk category with reported rates of distant metastasis up to 27% [4, 6, 7].

There is ongoing controversy regarding the extent of surgery and the need for adjuvant radioactive iodine (RAI) therapy based on the number of foci of angioinvasion in EAFTC. A few studies demonstrate that tumours with < 4 foci of angioinvasion are at low risk of distant metastasis and suggest no benefit from adjuvant RAI therapy [8–10]. Other studies, however, show that any number of angioinvasive foci pose a significant increased risk of distant metastasis [6, 11, 12]. This controversy is reflected in differing recommendations from experts, with the American Thyroid Association and Japanese Thyroid Guidelines classifying EAFTC with < 4 foci of angioinvasion as low risk tumours, adequately managed with a hemithyroidectomy [13, 14]. In contrast, European guidelines recommend a total thyroidectomy and adjuvant radioactive iodine therapy for all angioinvasive FTC [15–17]. Defining the optimal surgical approach and appropriate use of adjuvant RAI therapy is critical in improving outcomes for patients with follicular thyroid carcinoma.

The aims of this study were: (a) to investigate distant metastasis-free survival (DMFS) and disease-specific survival (DSS) in patients with follicular thyroid carcinoma, specifically within angioinvasive tumours (EAFTC) when stratified by the number of foci of angioinvasion; and (b) to investigate factors predicting distant metastasis and survival.

Methods

Patients

A retrospective cohort study of patients from 1990–2018 with a diagnosis of follicular thyroid carcinoma was conducted from a prospectively maintained surgical database. Variables of interest obtained from the database were age, gender, initial surgery performed (total or hemithyroidectomy), adjuvant radioactive iodine (RAI) therapy, histological characteristics (tumour size, the presence and extent of capsular invasion, number of foci and extent of angioinvasion), duration of follow-up, time when distant metastasis occurred and death due to thyroid carcinoma. If the presence or number of foci of vascular space invasion was not available in the database, the original glass slides were reviewed to provide this information. Death due to thyroid carcinoma data was obtained through data linkage with the New South Wales registry of births and deaths. The cohort was initially divided into three groups, MIFTC, EAFTC and WIFTC based on histological characteristics. The EAFTC group was further divided into two: < 4 foci and ≥ 4 foci of angioinvasion.

Statistical analysis

The groups were initially described using simple descriptive statistics in the form of mean ($\pm$ standard deviation), median ($\pm$ interquartile range) depending on distribution. Comparison between groups was then performed using Students t test, Mann–Whitney U or ANOVA for continuous variables depending on distribution and Chi-square testing for categorical variables. Outcomes of interest were distant metastasis-free survival (DMFS) and disease-specific survival (DSS). The outcomes of DMFS and disease DSS were analysed using survival analysis with Kaplan–Meier survival estimates and Cox proportional hazard modelling. Entry time was defined as time of surgery, with time to distant metastasis (DMFS) and death from disease (DSS) as end points. Patients who were lost to

follow up were censored. Statistical analysis was performed using STATA© software.

Histological definition of angioinvasion

Angioinvasion is defined in the 2017 WHO classification of endocrine tumours as the invasion of vessels within the capsule or beyond with intravascular tumour attached to the vessel wall, mixed with fibrin or covered with endothelium [4]. Angioinvasion should not be reported if tumour is simply pushing against a vessel wall or present as tumour thrombi floating within a vessel [4]. We defined FTC according to the 2017 WHO classification of thyroid tumours [4]. Tumours are classified as MIFTC if only limited capsular invasion is present. Grossly encapsulated angioinvasive follicular thyroid tumours (EAFTC) were further stratified into < 4 foci and ≥ 4 foci of angioinvasion. WIFTC was defined as tumours showing widespread invasion through the thyroid parenchyma, or extrathyroidal tissues [4]. We excluded Hurthle/Oncocytic cell carcinoma and follicular variant of papillary thyroid carcinoma, as these are recognised as distinctly different [4, 18]. Cases were reviewed by experienced pathologists [AG and JT].

Results

Overall

Between January 1990 and December 2018, 318 patients were identified with complete follow-up data available for 295. After excluding three for different histological diagnoses (two anaplastic, one papillary thyroid carcinoma), a total of 292 patients were eligible for analysis. Based on histological criteria, there were 139 (MIFTC) minimally invasive, 141 (EAFTC) encapsulated angioinvasive and 12 (WIFTC) widely invasive tumours. The baseline characteristics, histopathological data, treatment and follow-up by group (MIFTC, EAFTC and WIFTC) are summarised in (Table 1). Overall, the mean age at presentation was 52 years with a female preponderance (70.6%) with no statistically significant differences between the groups ($p = 0.067$ and $p = 0.517$, respectively). Tumour size increased significantly in more aggressive tumours ($p < 0.001$) with mean sizes of 2.6 cm, 3.6 cm and 4.2 cm in MIFTC, EAFTC and WIFTC, respectively**.** In more aggressive tumours, patients ($p < 0.001$) were significantly more likely to be managed with a total thyroidectomy (81.9%, 98.6% and 100%) and receive adjuvant RAI therapy (73.5%, 91.5% and 100%) in MIFTC, EAFTC and WIFTC, respectively (Table 1). Patients were followed up for a median of 6.2 years with no significant differences between the groups ($p = 0.79$). During this period of follow-up, overall distant metastasis-free survival was 89.4%. A significant progressive difference ($p < 0.001$) in distant metastasis (3.6%, 14.2%, 50%) and disease-specific death (2.9%, 7.8%, 33%) was observed in more aggressive tumours (MIFTC, EAFTC and WIFTC, respectively**;** Table 1).

Encapsulated angioinvasive follicular thyroid carcinoma

Overall, there were 95 encapsulated angioinvasive tumours with < 4 foci and 46 with extensive (≥ 4 foci)

Table 1 Overall descriptive data of 292 patients with follicular thyroid carcinoma by histology type (minimally invasive, encapsulated angioinvasive and widely invasive follicular thyroid carcinoma)

	MIFTC (n = 139)	EAFTC (n = 141)	WIFTC (n = 12)	p value
Age (years)*	48 ± 16	49 ± 19	60 ± 15	0.06
Gender (M:F)	42:96	38:104	5:7	0.49
Tumour Size (cm)*	2.6 ± 1.5	3.6 ± 2.2	4.2 ± 1.7	< 0.001[a]
Total thyroidectomy	113 (81.3%)	140 (99.3%)	12 (100%)	< 0.001[b]
RAI therapy	83 (59.7%)	119 (84.4%)	12 (100%)	< 0.001[b]
Distant metastasis (Recurrence)	5 (3.6%)	20 (14.2%)	6 (50%)	< 0.001[c]
Disease-specific death*	4 (2.9%)	11 (7.8%)	4 (33.3%)	< 0.001[c]
Follow-up (months)[+]	75 ± 87	71 ± 87	77 ± 73	0.62 [d]

[*]Due to thyroid carcinoma

[*]Mean ± sd

[+]Median ± IQR

[a]ANOVA

[b]X^2 test

[c]Fisher's exact test

[d]Kruskal–Wallis test

angioinvasion. Tumours were significantly larger ($p = 0.005$) at 4.3 cm vs 3.2 cm in extensive angioinvasion ($\geq$ 4 foci) (Table 2). Extensive angioinvasion significantly ($p < 0.001$) increased the risk of distant metastasis (30.4%) compared to angioinvasive tumours with < 4 foci (6.3%). Disease-specific survival decreased significantly ($p = 0.008$) in extensive angioinvasion (82.6%) compared to tumours with < 4 foci of angioinvasion (96.8%) (Table 2). There were no statistically significant differences in follow-up time, surgical treatment, age or gender between the two groups (Table 2). A large proportion of patients (91%) in both groups received adjuvant RAI therapy, respectively.

Distant metastasis and disease-specific survival

Overall, distant metastasis-free survival (DMFS) over a median follow-up of 6.2 years was 96.4%, 85.8% and 50% in MIFTC, EAFTC and WIFTC, respectively (Table 1). Six (19.4%) patients presented prior to surgery with distant metastasis. After initial adjuvant RAI therapy, an additional ten (32.2%) patients were diagnosed with distant metastasis. No patient with MIFTC had metastasis at presentation or after initial RAI therapy; in comparison, 50% of angioinvasive tumours with < 4 foci had distant metastasis detected either prior to surgery or after initial RAI therapy (Table 3). The most common site of metastasis was bone followed by lung. The overall DMFS in EAFTC tumours with < 4 foci was 93.7% and 69.6% in those with $\geq$ 4 foci of angioinvasion. When patients with distant metastasis at presentation were excluded, the long-term risk of distant metastasis in EAFTC with < 4 foci was 2.1% compared to 9.8% in tumours $\geq$ 4 foci of angioinvasion ($p = 0.015$) (Table 2). Log-rank testing of WIFTC tumours for distant metastasis stratified by angioinvasion (83.3% positive) was not significant ($p = 0.489$). Further

Table 2 Descriptive data of encapsulated angioinvasive follicular thyroid carcinoma (EAFTC) stratified by foci of angioinvasion

	Encapsulated angioinvasive follicular thyroid carcinoma (EAFTC)		
	< 4 Foci (n = 95)	$\geq$ 4 Foci (n = 46)	$p$value
Age (years)*	47 ± 18	53 ± 19	0.08[a]
Gender (M:F)	23:72	15:32	0.33[b]
Total thyroidectomy	93 (97.9%)	46 (100%)	0.32[b]
RAI therapy	77 (91.7%)	42 (91.3%)	0.94[b]
Tumour size (cm)*	3.2 ± 2.0	4.3 ± 2.4	0.005[a]
Capsular invasion	69 (72.3%)	36 (87.8%)	0.61[b]
Overall distant metastasis	6 (6.3%)	14 (30.4%)	< 0.001[b]
Distant metastasis during follow-up	2 (2.1%)	6 (13%)	0.015[d]
Disease-specific survival	96.8%	82.6%	0.003[b]
Follow-up (months)[+]	69 ± 74	84 ± 91	0.14[c]

[&]Death due to thyroid carcinoma

[*]Mean ± sd

[+]Median ± IQR

[a]t-test

[b]X^2 test

[c]Mann–Whitney U

[d]Fisher exact test

Table 3 Distant metastasis for histological subtype based on timing in relation to surgery

	MIFTC	EAFTC < 4 foci	EAFC $\geq$ 4 foci	WIFTC	Total
Timing metastasis detected	n (%)				
Prior to surgery	0 (0%)	1 (16.7%)	4 (28.6%)	1 (16.7%)	6 (19.4%)
After first RAI therapy	0 (0%)	3 (50%)	4 (28.6%)	3 (50%)	10 (32.2%)
During follow-up (> 12 m)	5 (100%)	2 (33.3%)	6 (42.8%)	2 (33.3%)	15 (48.4%)
Total	5	6	14	6	31

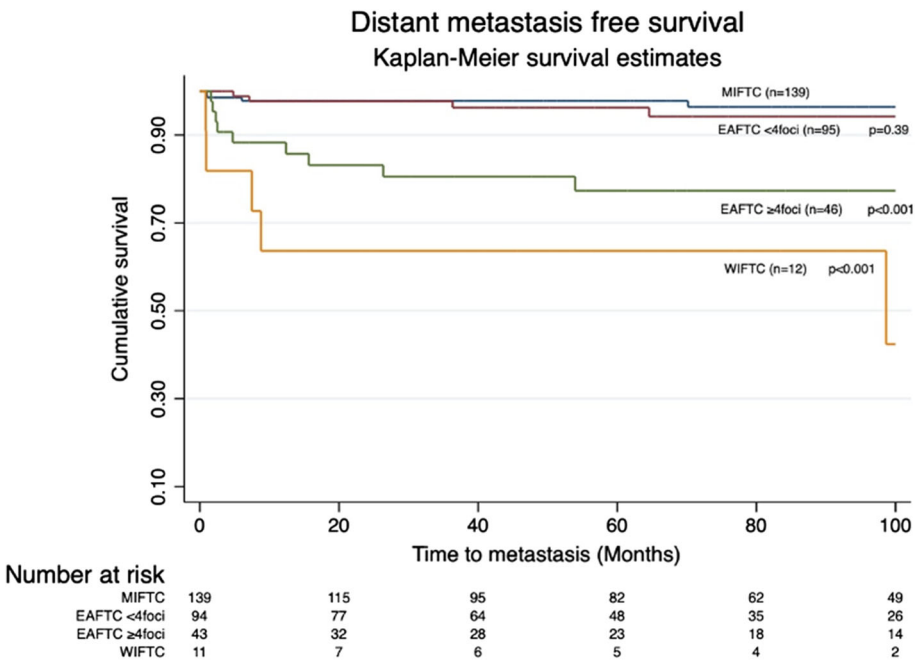

Fig. 1 Kaplan–Meier survival estimates for time to distant metastasis by FTC histological subtype

log-rank testing for distant metastasis and disease-specific survival in both MIFTC ($p = 0.166$) and EAFTC ($p = 0.10$) groups that received RAI therapy was also not significant.

Kaplan–Meier survival estimates in Fig. 1 show encapsulated angioinvasive tumours have a poorer prognosis with recurrence rates of extensive angioinvasive (≥ 4 foci) tumours approximating WIFTC. Disease-specific survival over 6.2 years was 97.1%, 92.2% and 66.7% in MIFTC, EAFTC and WIFTC, respectively. Disease-specific survival in EAFTC decreased significantly when stratified by foci of angioinvasion Kaplan–Meier survival estimates for DSS in Fig. 2 show significantly decreased survival in EAFTC ≥ 4 foci and WIFTC. Kaplan–Meier survival estimates for distant metastasis by excluding patients with metastasis detected after the first RAI therapy are included in supplementary online material.

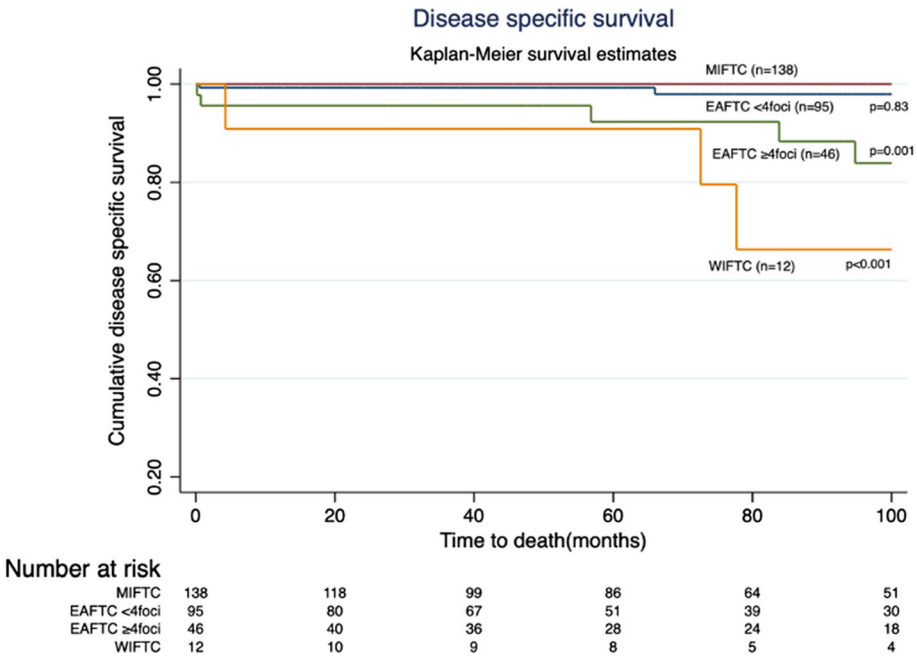

Fig. 2 Kaplan–Meier survival estimates for disease-specific survival by FTC histological subtype

Factors affecting distant metastasis in FTC

Prognostic factors were analysed using Cox proportional hazard regression. Multivariable regression shows that age ≥ 50 y (HR 4.24, 95%CI 1.56–11.57) and tumour size (HR 1.27/cm, 95% CI 1.04–1.53) increased the risk of distant metastasis significantly Table 4. When a tumour size cutoff was evaluated, tumours greater than 3cms had a significantly threefold increased risk of distant metastasis (HR = 3.57, $p = 0.001$). The mean age at presentation with distant metastasis was 60 y compared to 47 y with non-metastatic disease. The risk of distant metastasis also increased significantly ($p = 0.004$ and < 0.001) in EAFTC with extensive (≥ 4 foci) angioinvasion (HR 5.89) and widely invasive tumours (HR 12.05), respectively. EAFTC tumours with < 4 foci were not at significant increased risk of distant metastasis compared to MIFTC Table 4.

Factors affecting disease-specific survival in FTC

Factors affecting DSS were analysed using Cox proportional hazard regression. Multivariable regression demonstrates age ≥ 50 y (HR 11.73, 95%CI 1.52–90.22, $p = 0.018$) and WIFTC histology subtype (HR 6.97, 95%CI 1.39–34.93, $p = 0.018$) were highly significant for increased risk of death due to metastatic follicular thyroid carcinoma (Table 5). Gender and tumour size were not significantly associated with survival outcomes in

Table 4 Univariable and multivariable Cox proportional hazard modelling for distant metastasis by age, tumour size(cm) and histological subtype

	Univariable			Multivariable		
	HR	95% CI	*p* value	HR	95% CI	*p* value
Age						
< 50	Ref			Ref		
≥ 50	5.2	1.96–13.79	0.001	4.24	1.56–11.57	0.005
Gender						
Male	ref			ref		
Female	0.56	0.26–1.21	0.142	0.76	0.34–1.67	0.49
Tumour size (cm)	1.46	1.25–1.69	< 0.001	1.27	1.04–1.53	0.014
Histology subtype						
MIFTC	Ref					
EAFTC < 4 foci	2.0	0.54–7.49	0.30	1.74	0.46–6.67	0.417
EAFTC ≥ 4 foci	10.3	3.31–31.97	< 0.001	5.89	1.77–19.61	0.004
WIFTC	25.4	7.10–91.05	< 0.001	12.05	3.03–47.93	< 0.001

Table 5 Univariable and multivariable Cox proportional hazard modelling for disease-specific survival by age, tumour size (cm) and histological subtype

	Univariable			Multivariable		
	HR	95% CI	*p* value	HR	95% CI	*p* value
Age						
< 50	Ref			Ref		
≥ 50	19.37	1.96–13.79	0.004	11.73	1.52–90.22	0.018
Gender						
Male	Ref			Ref		
Female	0.36	0.14–0.93	0.035	0.55	0.19–1.53	0.252
Tumour size (cm)	1.38	1.12–1.69	0.002	1.16	0.92–1.47	0.218
Histology subtype						
MIFTC	Ref					
EAFTC < 4 foci	1.07	0.18–6.41	0.94	0.97	0.16–5.94	0.975
EAFTC ≥ 4 foci	7.36	1.95–27.79	0.003	3.69	0.89–15.26	0.071
WIFTC	15.87	3.55–71.0	< 0.001	6.97	1.39–34.93	0.018

multivariable analysis. The presence of distant metastasis alone predicts a significant risk of dying of disease (HR 23.82, 95%CI 6.53–86.88, $p < 0.001$). When cases without metastasis were analysed with Cox proportional hazard regression, both age and WIFTC histology subtype were not significant ($p = 1.00$).

Discussion

In this study, we show that encapsulated angioinvasive follicular thyroid carcinoma (EAFTC) is a distinct higher risk category with increased rates of distant metastasis (14.2%) in comparison to MIFTC with capsular invasion only (3.6%). The degree of angioinvasion correlates significantly with the risk of distant metastasis and overall survival. When stratified by number of foci, EAFTC with ≥ 4 foci of angioinvasion is at high risk (30.4%) of distant metastasis and decreased survival. EAFTC with < 4 foci of angioinvasion, however, has an intermediate risk of distant metastasis (6.3%) compared to minimally invasive follicular carcinoma (MIFTC) with capsular invasion alone (3.6%). In addition, we show that the risk of distant metastasis occurring after initial therapy in patients with EAFTC tumours with < 4 foci during long-term follow-up (> 12 months) is very low (2.1%). Other studies also demonstrate EAFTC tumours represent a distinct higher risk category with higher rates of distant metastasis and reduced survival [6, 8, 10, 15, 19–23]. Two recent papers which stratified FTC by the number of foci show all FTC tumours with ≥ 2 foci angioinvasion are of increased risk of metastasis and reduced survival [12, 24]. A large proportion of patients ($> 50\%$) in the two recent studies [12, 24] did not receive RAI therapy, however, which may possibly account for the differences in outcomes in comparison to our study. Additionally, in our study, 83.3% ($n = 10$) of WIFTC tumours with vascular invasion were not significantly ($p = 0.498$) at increased risk of distant metastasis compared to WIFTC without angioinvasion. This finding is in contrast to a recent study by Ito et al. where WIFTC tumours with angioinvasion were at increased risk of distant metastasis and reduced survival, whereas WIFTC tumours without angioinvasion were prognostically similiar to MIFTC [12]. This difference in finding may be related to the low numbers of WIFTC patients in our cohort with the majority having angioinvasion.

The controversy of whether therapy with a total thyroidectomy and adjuvant RAI therapy improves outcomes in EAFTC tumours with < 4 foci of angioinvasion and has only been addressed by a few studies. Ito et al. followed 242 patients with EAFTC tumours < 4 foci of angioinvasion treated with a hemithyroidectomy without metastasis at time of surgery and reported a 5% risk of distant metastasis over a period of 9 years [8]. Lee et al. and Sugino et al. compared MIFTCs ($n = 96$ and 223, respectively) treated with a hemithyroidectomy to patients receiving a total thyroidectomy with adjuvant RAI therapy and found no significant difference in terms of distant metastasis (3% and 6%, respectively) [25, 26]. However, neither study stratified by foci of angioinvasion and it is unclear if the metastasis occurred in those with extensive vascular invasion only [25, 26]. Our results may suggest that lower rates of distant metastasis (2.1%) during follow-up could be attributed to the increased use of adjuvant RAI therapy compared to the higher rates of metastasis reported in the literature.

Another distinctive finding in this study is that a substantial proportion of metastasis in angioinvasive follicular thyroid carcinomas is present at the time of initial presentation. Of all patients who developed distant metastasis, in $> 50\%$, these were detected by the first RAI ablative dose. Other studies of angioinvasive follicular thyroid carcinoma report similar rates of distant metastasis [8, 9, 27]. Performing a total thyroidectomy upfront allows for determining biochemical response with serum thyroglobulin levels [28–30]. Thyroglobulin levels in differentiated thyroid carcinoma post-thyroidectomy are a highly sensitive indicator of potential distant metastasis which permits earlier detection and treatment with RAI therapy [31–34]. Elevated thyroglobulin levels post-thyroidectomy predicts an increased risk of developing structural recurrence in up to 20% of cases [13]. The earlier detection of distant metastasis treatment with adjuvant RAI therapy improves survival outcomes for patients with differentiated thyroid carcinoma [35, 36].

Additionally, this study shows increasing age at diagnosis significantly increases the risk of distant metastasis and death from follicular thyroid carcinoma. This finding is consistent with other studies reporting significantly higher rates of distant metastasis in patients over the age of 45 [19, 37, 38]. In contrast, other studies report no significant difference [39]. It is unclear why there is a discrepancy, and however, the numbers of distant metastasis is smaller in the studies, which show no difference only on multivariate analysis and may reflect significant confounding due to angioinvasion as a stronger predictor of distant metastasis when included in statistical analysis. In this study, increasing tumour size significantly increased the risk of distant metastasis but not death from thyroid carcinoma**.** This finding is supported by studies showing increasing tumour size in follicular thyroid carcinomas correlating with an increased risk of distant metastasis [8, 9, 19]. In keeping with increasing tumour size as a poor prognostic factor, both the American Thyroid Association guidelines and European guidelines typically recommend a

total thyroidectomy in minimally invasive follicular thyroid carcinomas ≥ 4 cm [13, 15]. These findings suggest that a total thyroidectomy and adjuvant RAI therapy may not be indicated in younger patients with small minimally invasive follicular thyroid carcinomas.

A limitation of this study is that a large proportion (90%) of patients with angioinvasive follicular thyroid carcinoma were treated with a total thyroidectomy and adjuvant RAI therapy. The large proportion of patients treated with adjuvant RAI therapy makes comparison between groups receiving RAI therapy difficult. Another limitation is lenvatinib, a tyrosine kinase inhibitor that became available in Australia since 2017; as our surgical database does not capture treatment with lenvatinib, this could potentially affect DSS times. Further studies comparing rates of distant metastasis and survival in EAFTC with < 4 foci managed with a hemithyroidectomy only with stringent histological criteria would be beneficial.

Conclusion

Encapsulated follicular thyroid carcinoma with < 4 foci of angioinvasion has an increased risk of metastasis, particularly in older patients and larger tumours. There is a role for a total thyroidectomy and adjuvant RAI therapy.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00268-022-06696-6.

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Conflict of interest The authors declare that there is no conflict of interest

Ethics approval Ethics approval was obtained from the local hospital ethics committee.

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